Theme:

Date: / /

Sat Sun Mon Tue wed Thu Fri

Chirality in Molecules Devoid of Chiral Centres

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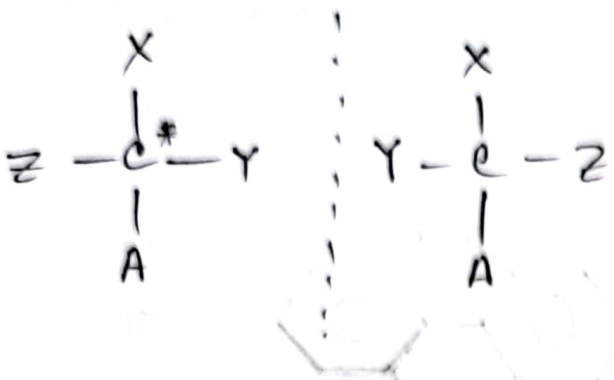
CamScanner

4th Chapter

Theme:

Date: 10/11/24

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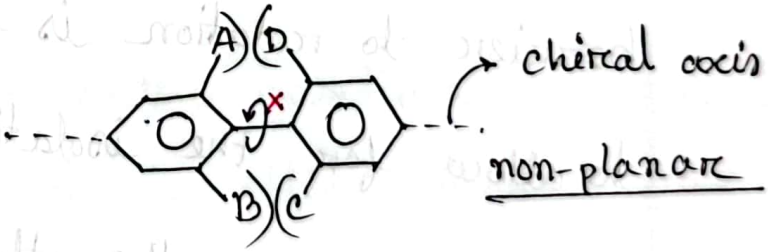


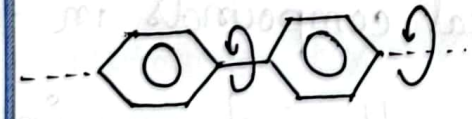
* When the mirrcore image of a molecule is non-superimposable to the molecule, that is called a chircal molecule.

1. Chiral centre
2. Chiral axis
3. Chiral plane
4. Molecular helicity

} 2+4 combination বা, অন্তত combination বা যেকোনো 1টি থাকলে molecule chiral হবে থাকে।

A necessary & sufficient condition for a molecule to be chiral is that molecule is non-superimposable to its mirrcore image.



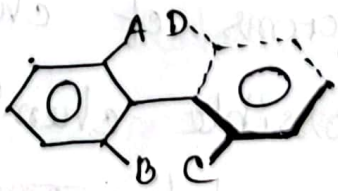


Planar

C-C rotation হয় (free) at R.T.

C-C free rotation হয় না steric hindrance থাকে A) (D B) (C

↓ minimization



(দুটো ring perpendicular হয়ে থাকে)

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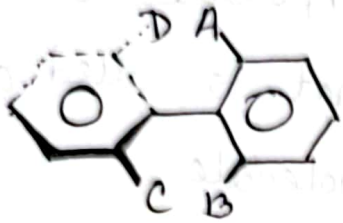
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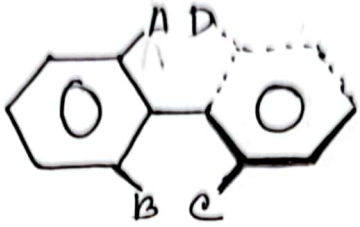
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Atrope Isomerism :

Atrope → Absence of rotation

(I)





(II)

Non-superimposable MI ⟶ Chiral molecule

↳ এখানে rotation hindered হচ্ছে না। Chiral axis আছে, free rotation এজন্য chiral axis নষ্ট না।

(I) ও (II) পরস্পরের atrope isomer.

Atrope isomers are stereoisomer resulting from hindered rotation about single bond, where the steric barrier to rotation is high (75 - 165 KJ/mole) enough to allow for the isolation of conformers.

They differ from the other chiral compounds in that they can be equilibrated thermally whereas in other forms of chirality isomerization is usually only possible chemically.

[তাপ দিয়ে rotation করলে chirality নষ্ট হয়]

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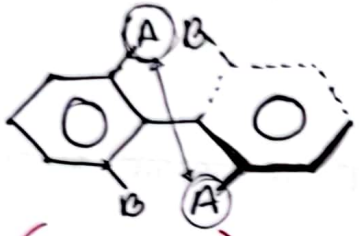
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Two conditions are necessary for diphenyl compound to exhibit optical activity.

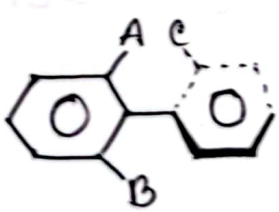
(1) Neither ring must have a vertical plane of symmetry.
(কোনো একটি ring-এ plane of symmetry থাকলেই molecule টি achiral হয়)
(কোনো symmetry না থাকলে molecule 'chiral' হয়)

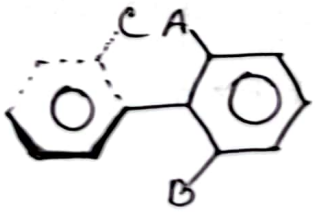


এখানে,
- Plane of symmetry নাই ✗
- Centre of symmetry নাই ✗
- প্রতি ring এ নিজস্ব plane of symmetry নাই ✗

Achiral
যেহেতু COS থাকাতে মনে হলেও আসলে 2 ring একই plane এ না থাকায় COS নেই।

(2) Two substituents in the ortho positions must have a large size so that steric effect prevent free rotation. Ex- Cl, Br, I, COOH, NO_2, SO_3H, CH_3





→ A, B, C behave as obstacle. The free rotation about the single bond is restricted.

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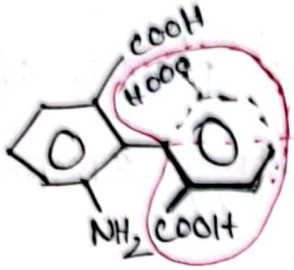
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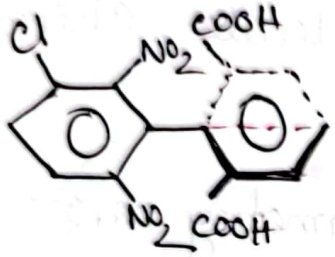
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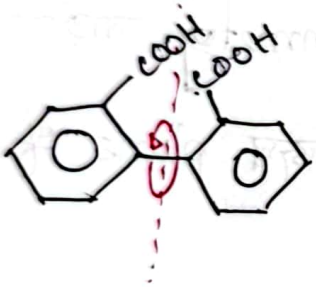
Sat Sun Mon Tue wed Thu Fri

Thus the two phenyl rings cannot be coplanars, remain perpendicular to each other.

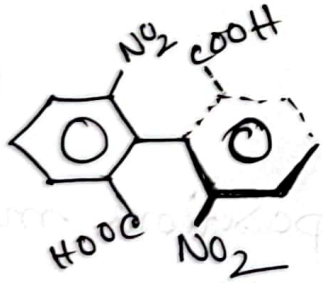
i) 

→ Plane of symmetry আছে → achiral

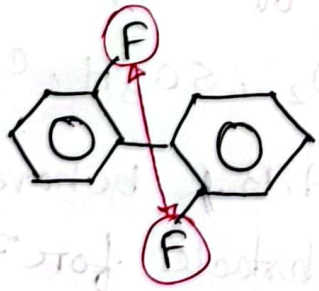
ii) 

iii) 

→ Temp দিলে কোনো না কোনো সময় এটা planar হয়ে rotate করে। তখন এটার মধ্যে P.O.S. থাকে। তাই → achiral

iv) 

→ Chiral → Centre of symmetry, plane of symmetry নাই।

v) 

→ 'F' ছোট তাই hindered না। Centre of symmetry আছে। → achiral.
F ↔ F

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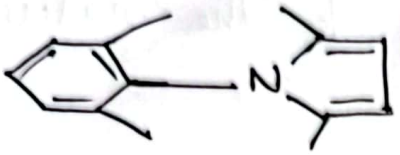
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Date: 12 / 11 / 24

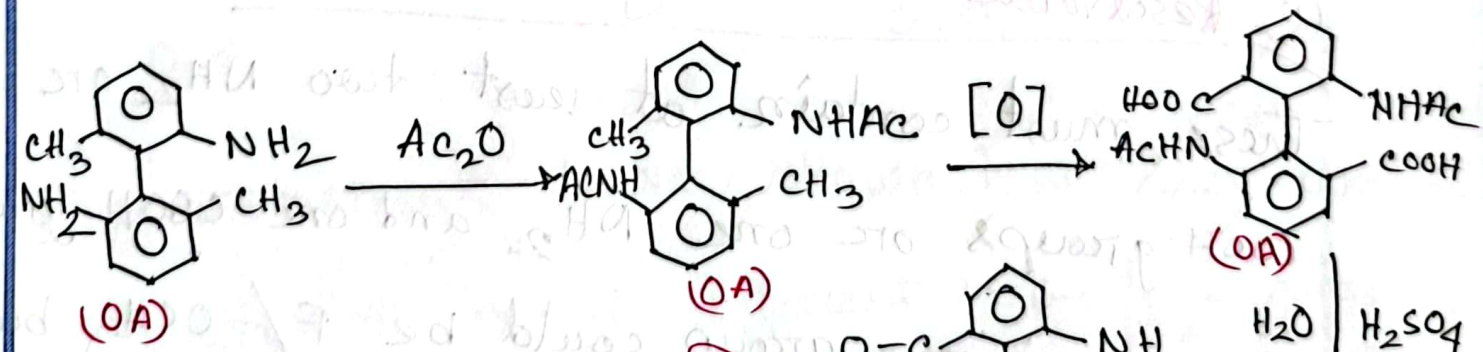
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Obstacle Theory :



Large groups had to be present in the ortho position to behave as obstacles to present free rotation about single bond joining two benzene rings. Thus two benzene rings cannot be coplanar and optically active biphenyl compounds can isolated rotation.

Evidence



Dilactum: Planar has a centre of symmetry. So, optically inactive.

Optically Inactive (OI)

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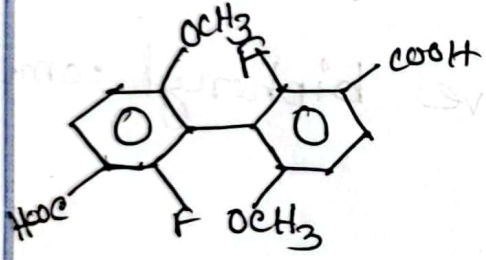
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2,2',6,6'- tetra substituted biphenyl compounds may be classified according to the nature of the substituents.

① Non-resolvable

These contain any of the groups (H, $-OCH_3$, F). The effective volumes of these groups are too small to prevent rotation.



$>2.9\,A^{\circ}$

Not resolvable

② Resolvable but easily racimized

These must contain at least two NH_2 or COOH groups or one NH_2 and one COOH group. The remaining group could be F/$-OCH_3$ but not (-H).

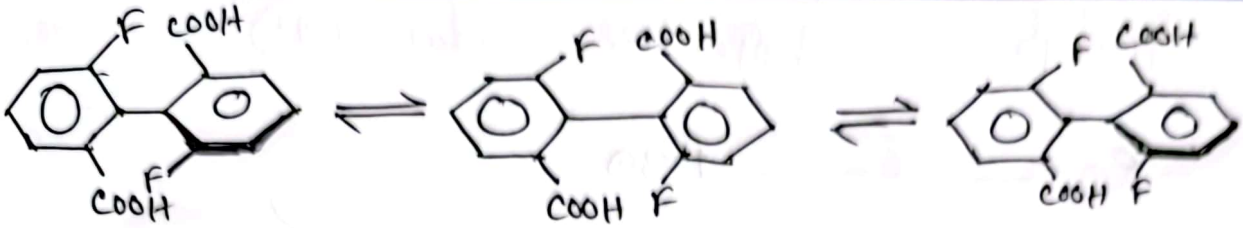
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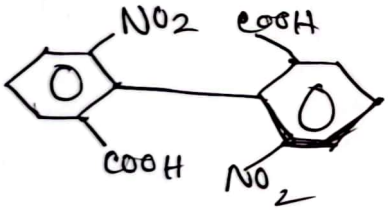
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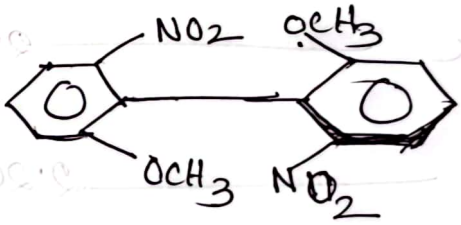
O.A. O.Inactive O.A.

(B) Not Racemisable at all

The biphenyl compounds contain at least two NO_2 groups. The remaining group may be any of $-OCH_3$, NH_2, COOH groups.



O.A.



O.A.

(C) Measurements have shown that the sum of the effective raddii ~~different~~ ~~of~~ the groups in ortho position in different ring has to be higher than 2.9 A°.

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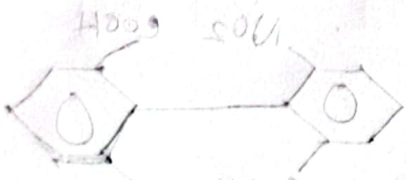
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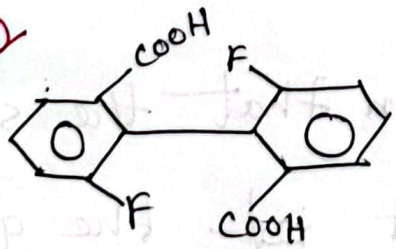
Group	Effective raddii (A°)
$-F$	1·39
$-H$	0·94
$-OH$	1·45
$-COOH$	1·73
$-CH_3$	1·56
$-Cl$	1·89
$-NO_2$	1·92
$-Br$	2·11
$-I$	2·20

ortho





(1)



F = 1·39

$-COOH = 1·73$

3·12 > 2·9 → optically active resolvable but easily racimized

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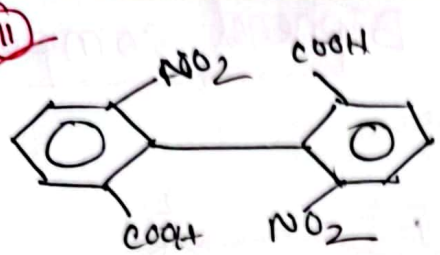
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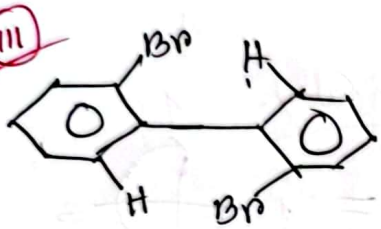
(II)



COOH – 1·73
NO2 – 1·92
3·65 > 2·9 A°

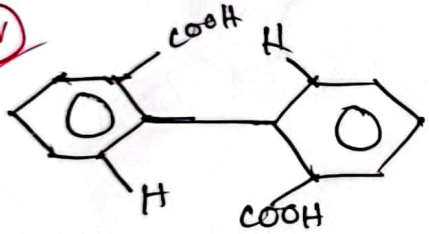
optically active

(III)



H – 0·04
Br – 2·11
3·05 > 2·9 → optically active easily racemized

(IV)



H – 0·04
COOH – 1·73
2·67 < 2·9

optically inactive

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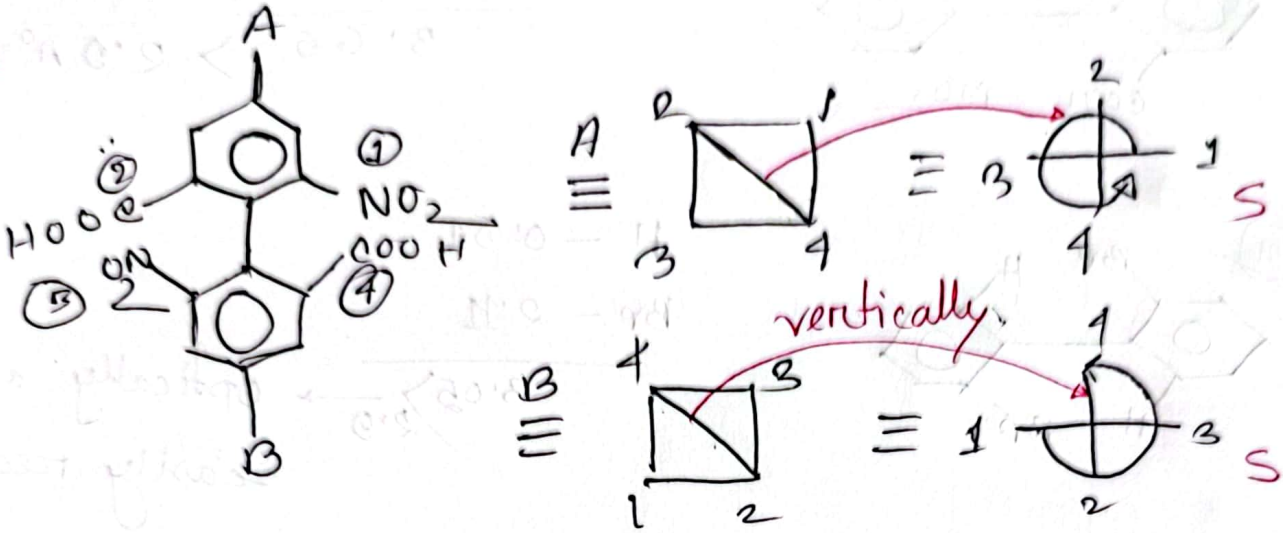
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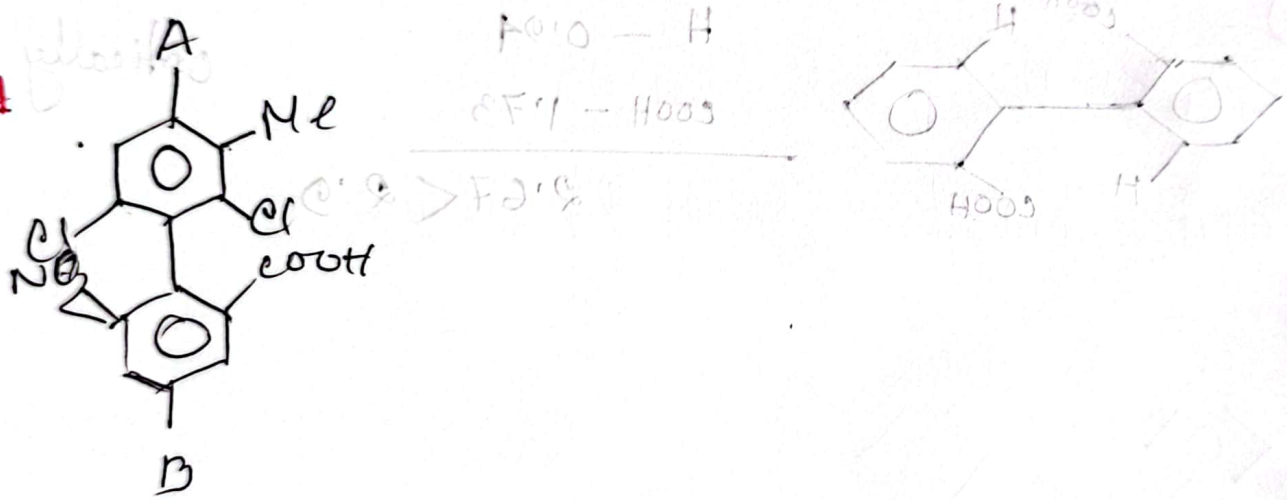
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Absolute Configuration of Biphenol compounds



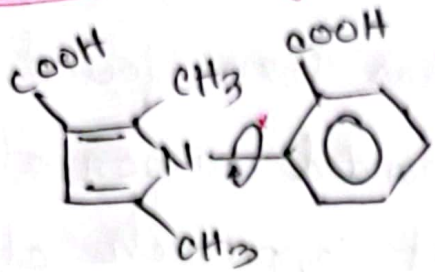


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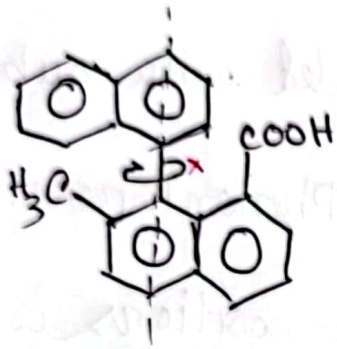
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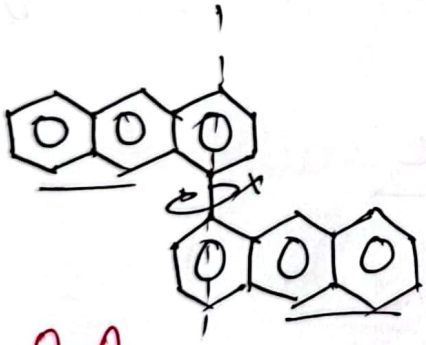
Optical Activity without Biphenyl compounds



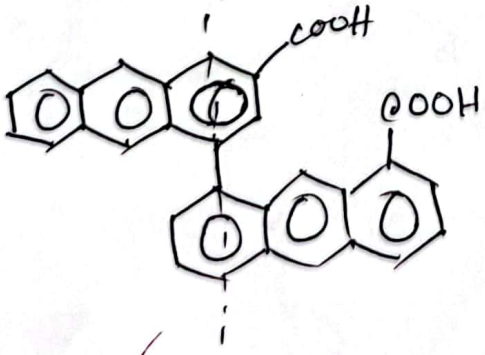
O.A.



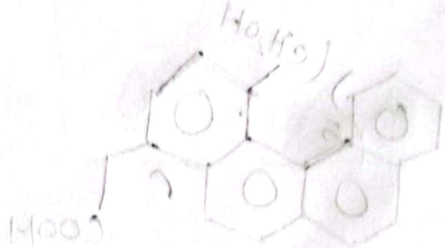
O.A.



O.A



O.A

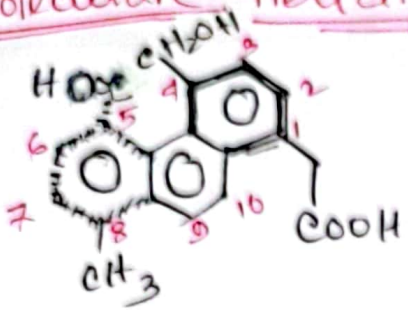


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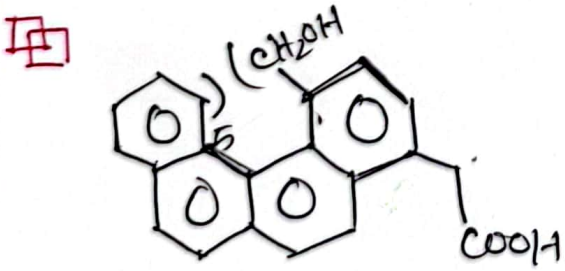
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Molecular Helicity / Overcrowding:



It has been found that non-bonded carbon atom cannot approach closer to each other than 3.0 A°. When th 1,5,8 positions of phenanthrene is overcrowded with substituents the molecule is not planare. Phenanthrene itself is planare. But if the Cis positions is occapied by long substituents there will not enough room.

Causing Steric Hindrance



O.A.

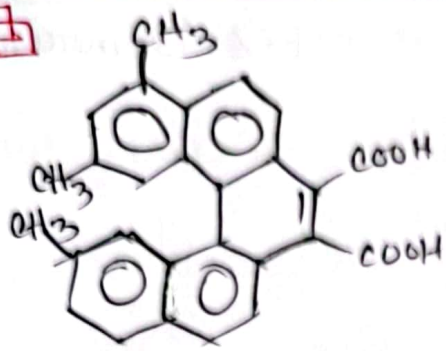
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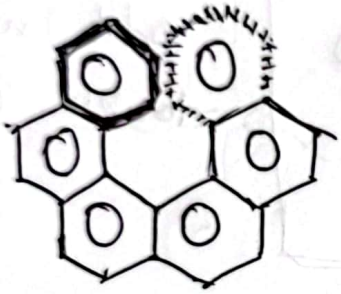
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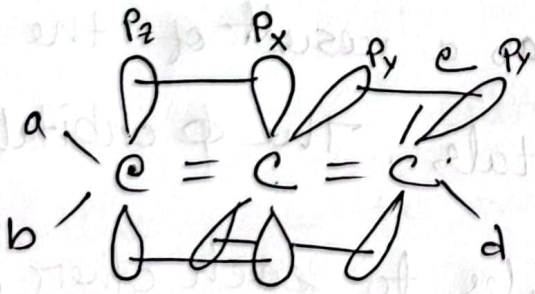


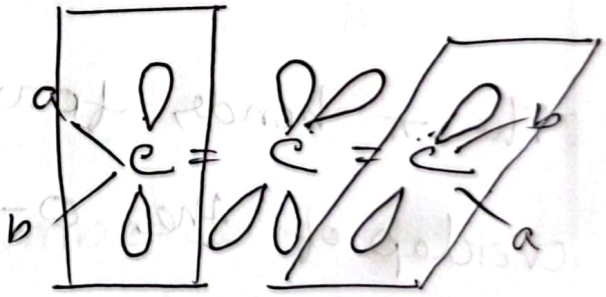
O.A.



O.A.

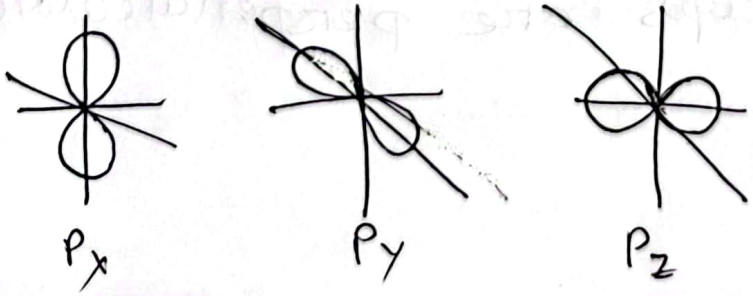
Allene Cumelene





$a \neq b$

$c \neq d$



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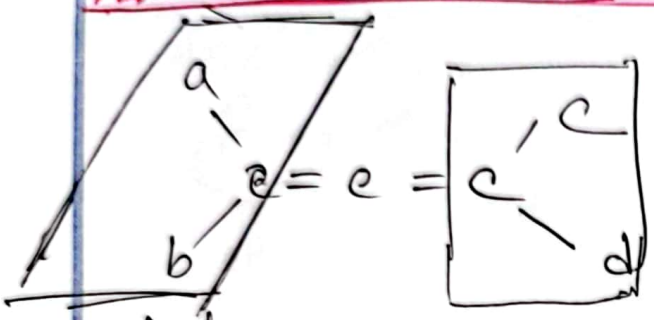
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Allene Stereochemistry:

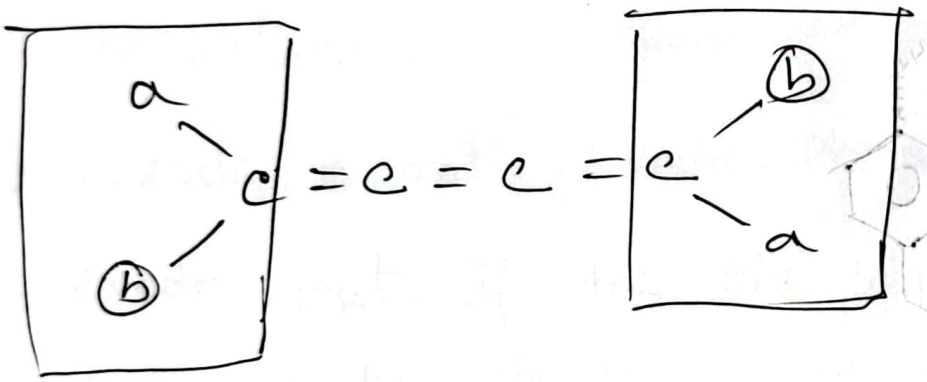
E-Z isomerism



Chiral Plane

O.A.

Even no. of double bond



odd no of double bond

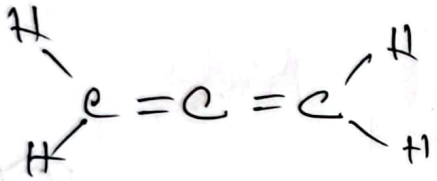
Optically inactive but have geomtrical isomerism.

The π bonds formed as a result of the overlap of the p-orbitals. The p orbitals must be at right angles to each other. The two methylene groups are perpendicular.

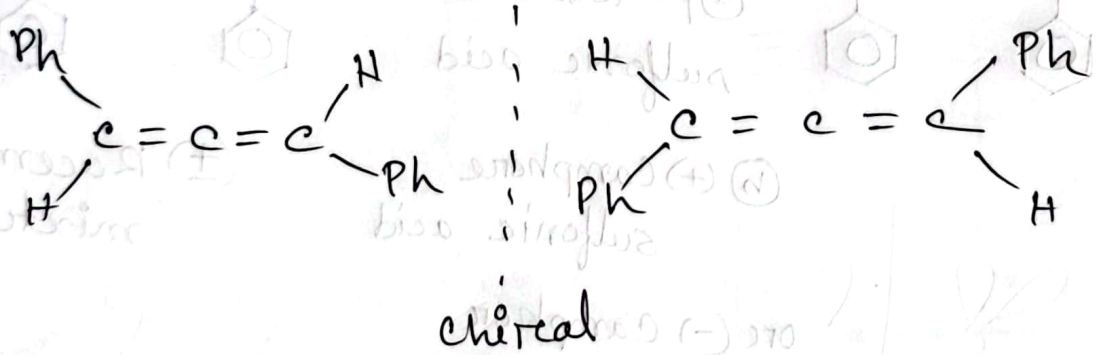
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The groups of a and b lies in the plane of the paper and the groups d and c in the perpendicular to the plane of the paper. This molecule does not possess a plane or centre of symmetry.

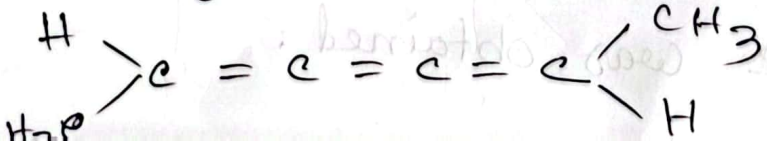


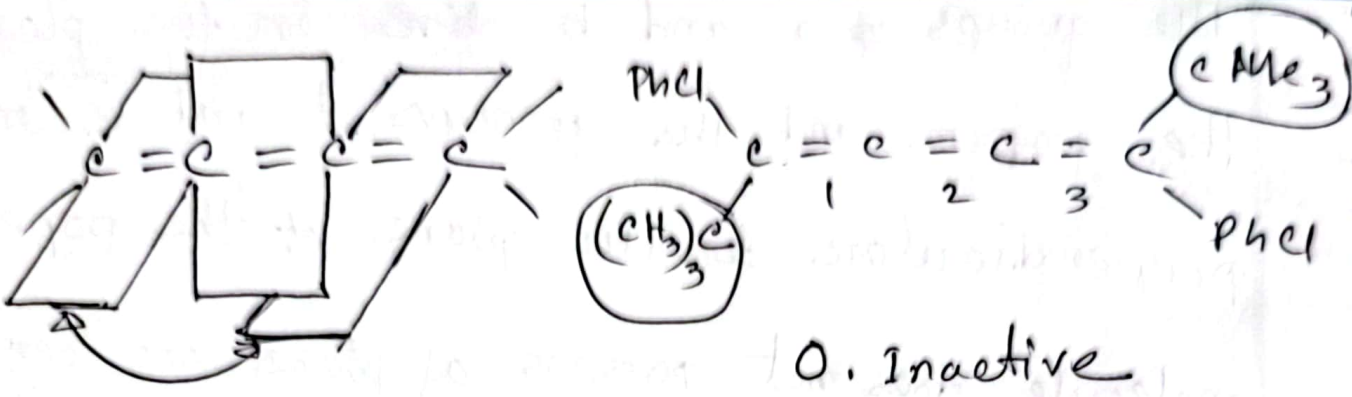
1,2 propadiene has plane of symmetry. Achiral, When different substituents are added, it is chiral.



chiral

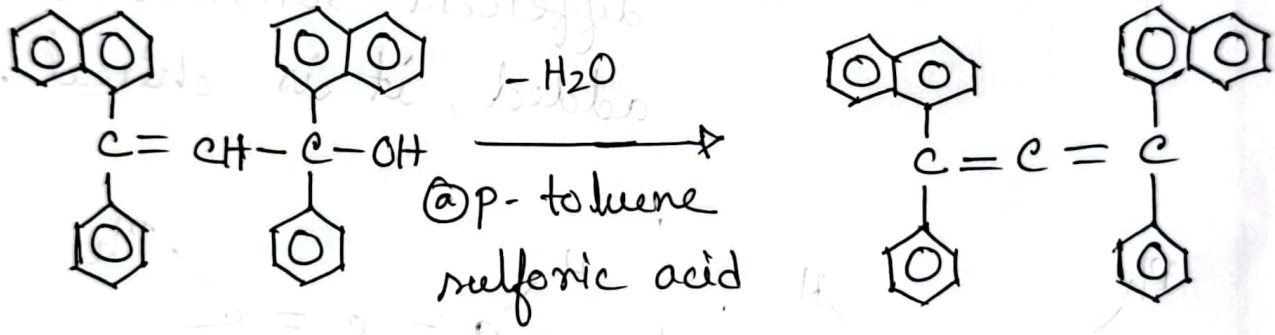
Alkenes with even numbered of cumulated double bonds are optically active if the groups present on the terminal carbons are different. But if the groups are same, then achiral.





O. Inactive trans.

Allene Synthesis

① 

ⓐ p-toluene sulfonic acid

ⓑ (+) Camphore sulfonic acid

or (−) Camphor sulphonic acid

(±) Racemic mixture

ⓐ When dehydration with an optically inactive dehydrating catalyst, p-toluene sulphonic acid is used, the racemic mixture of the alkene derivative was obtained.

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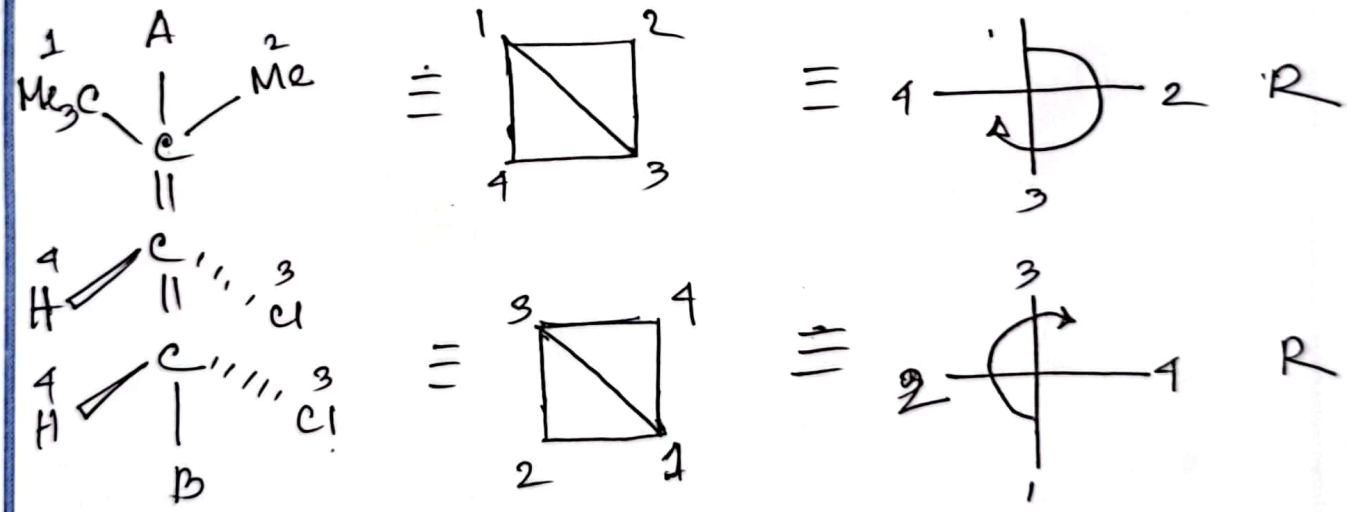
(b) But when the alcohol was boiled with 1% benzene solution of (+) camphor sulphonic acid, a dextro-rotatory [(+)] alkene was obtained.

(2) $PhC \equiv CCH_2Ph \xrightarrow[\text{Alkaloid}]{Al_2O_3} PhCH=C=CHPh$

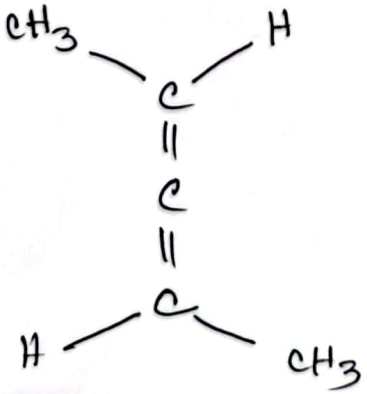
→ (−) Alkene

Quinine → (+) "

Absolute Configuration of Alkene



(*)



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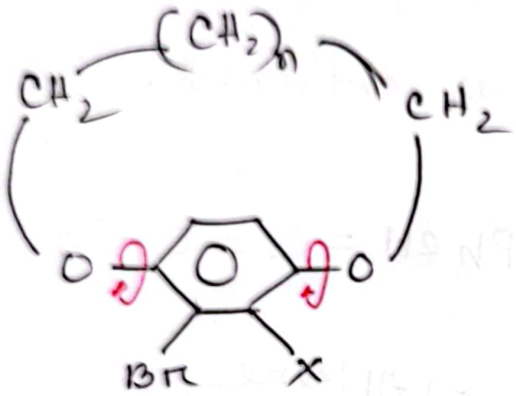
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Ansa type Compound



$n = 1$ ... $CH_2 = 3$ O.A.

$n = 2$ O.A

$n = 3$ O.I.

Optically active due to chiral plane

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